

Taking AI-DLC to 60% adoption across data-driven domains

Generic AI-DLC kits don't fit data work. I adapted them for data science, brought them to Databricks, and I'm taking them to data analysis, data engineering and data science teams.

Skill

Agent

Done

Planned or ongoing

OBJECTIVE

60% AI-DLC adoption by end of 2026

AI-DLC IN DATA-DRIVEN DOMAINS

Data analysis

Data engineering

Data science

6

sessions delivered: 3 knowledge sharing, 3 Pixel training

2

AIDLC O&E coach sessions planned for Oct 1–2

4

data science skill packs, shared in the AI Shark repo

2

agents: AI Ops and code reviewer

4,000+

people enabled on AIDLC through AI Sharkathons

⚠ THE GAP

Generic kits don't fit data work

Spec Kit and similar kits are built for app development. Data science adds steps they don't cover:

Data profiling

Leakage checks

Feature selection

Model evaluation

Drift monitoring

No spec-driven tool on Databricks

Databricks, where our data teams build, had no spec-driven development tool.

Teams need to see it work

Adoption needs proof on real work, not toy examples, so the member risk build is the live demo.

✓ CONTRIBUTION: SKILLS AND AGENTS

Shared: AI Shark repo

→

Integrated: UHC catalog skills repo

Token economics skills

Token budgets and lean prompts

Context management skills

Right context, loaded on demand

Feature selection skills

Ablation, SHAP, leakage checks

Guardrails skills

PHI, security, leakage checks

AI Ops agents

Drift triage, CRs, reruns

Code reviewer agent

Pass, fix or block on every PR

Spec Kit on Databricks

Spec-driven development running inside Databricks, on our model endpoints and under Unity Catalog governance.

👥 DRIVING ADOPTION

Teams adopting AI-DLC, Ramesh's org

+3 TEAMS IN SEPT

4

3

2

1

0

0

3 teams

Aug

Sept

Biweekly knowledge sharing, Ramesh's org

3 sessions demoing the member risk work

DONE

Pixel mandatory training coach

3 sessions at Kalyani Magnum, Bangalore

DONE

AIDLC O&E coach

Sessions on Oct 1 and Oct 2

PLANNED

Sharkathon

Collaborating with the AI Shark team

ONGOING

NEXT 90 DAYS

Path to 60%

OCT 1–2

AIDLC O&E coach sessions

→

EVERY TWO WEEKS

Knowledge sharing in Ramesh's org

→

NEXT

Feature creation trial: vibe coding vs AI-DLC

→

DEC 2026

60% AI-DLC adoption in data-driven domains

Each pipeline stage is built through the same harness

Feature creation is done end to end with Spec Kit. Selection, training, inference and MLOps follow the same path and are in progress.

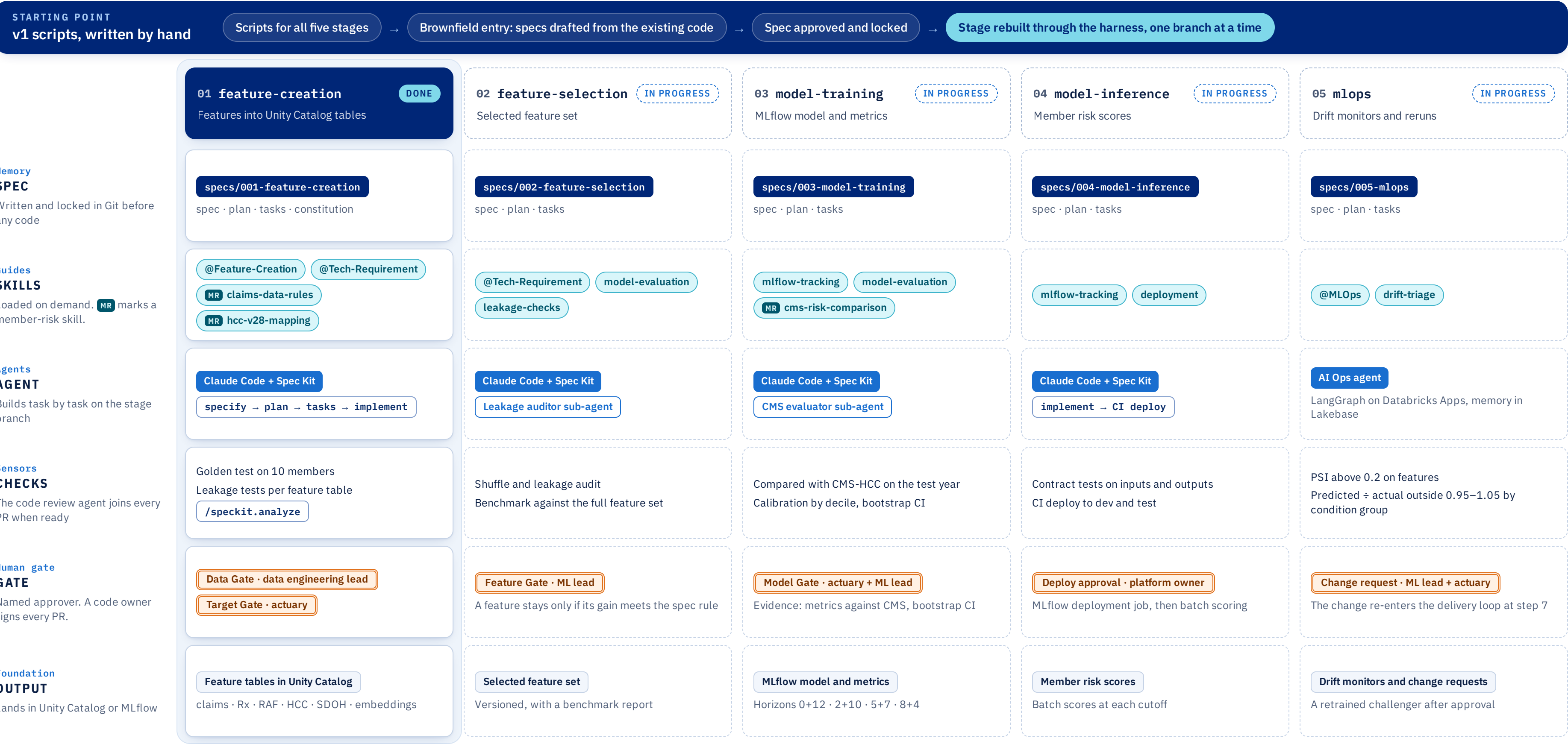
- Agent

Skill

Spec Kit command
- Git

Human gate

In progress



Same path for every stage: one branch, a locked spec, an agent build, checks, a code-owner PR approval, then the stage gate. A change reruns only from the earliest stage it touches.

Every ML change ships through one governed, agent-assisted loop

Agents do the work, humans own the gates, and the spec stays the source of truth from data to production.

Agent

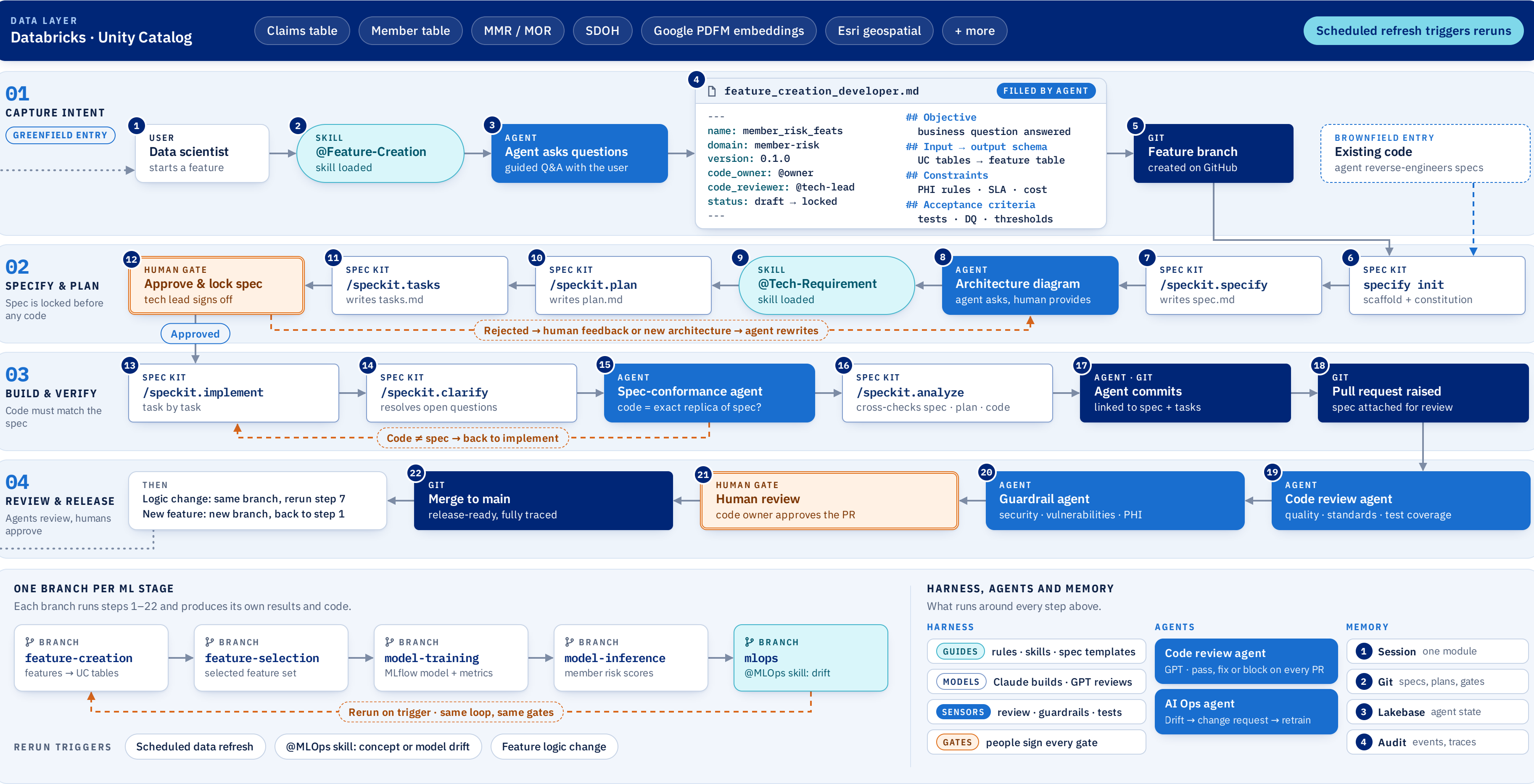
Skill

Spec Kit command

Git action

Human gate

Reject / rerun loop



What's reusable, what's member-risk only, and what I built

The global AI-DLC package installs into any Databricks data science repo. The member-risk pack is the worked example on top of it.

AI-READY DATA SCIENCE REPO

Each stage is a self-contained module. Agents read the rules and the spec before they touch code.

member-risk-model/		
└─ AGENTS.md	Rules every agent reads first	GLOBAL MEMORY
└─ CODEOWNERS	One owner group per stage	LOCKED
└─ databricks.yml	Bundle: dev · test · prod	GLOBAL
└─ .specify/	Spec Kit	
└─ memory/constitution.md	10 rules: no leakage, no PHI	GLOBAL MEMORY LOCKED
└─ templates/	Spec, plan, tasks, feature	GLOBAL
└─ skills/		
└─ global/	From the AI-DLC package	GLOBAL
└─ member-risk/	Claims, HCC V28, CMS	MEMBER RISK
└─ specs/NNN-change/	Spec · plan · tasks · approval	MEMORY LOCKED
└─ gates/	Signed gate records	MEMORY LOCKED
└─ common/	mrs_formulas, IO, MLflow helpers	MEMBER RISK
└─ pipelines/pipeline.yaml	Stage order, contracts, triggers	
└─ stages/	One self-contained module each	
└─ 01_feature_creation/		DONE
└─ 02_feature_selection/		IN PROGRESS
└─ 03_model_training/		IN PROGRESS
└─ 04_model_inference/		IN PROGRESS
└─ 05_mlops/		IN PROGRESS
each stage: config/ · contract.yaml · src/ · tests/ · AGENTS.md · job.yaml		
└─ .github/workflows/	CI, AI code review, deploy	GLOBAL
└─ tools/code_review_agent/	The aidlc_review() package	GLOBAL

SKILLS AND ASSETS

GLOBAL · AI-DLC PACKAGE		Any Databricks data science team
SPEC KIT COMMANDS · /SPECKIT.		
specify	clarify	plan tasks analyze implement
BUILD SKILLS		
@Feature-Creation	@Tech-Requirement	ds-data-profiling
feature-store-patterns	config-driven-module	leakage-checks
mlflow-tracking	model-evaluation	
REVIEW AND OPERATE		
request-code-review	gate-review	@MLOps drift-triage
SUB-AGENTS		
spec reviewer	leakage auditor	data validator test writer
TEMPLATES AND PLATFORM TABLES		
data spec	feature spec	target definition model card
evaluation protocol	monitoring plan	change request aidlc_gates
aidlc_events		

MEMBER RISK PACK		This model only
DOMAIN SKILLS		
claims-data-rules	hcc-v28-mapping	cms-risk-comparison
Claims lag, reversals and member-months · V24 to V28 HCC codes · the test against CMS-HCC		
FEATURE RULES		
MMR / MOR	SDOH	PDFM embeddings Esri geospatial
SPEC, CODE AND TESTS		
MRS-001 · 17 modules	mrs_formulas F1–F8	Golden test: 10 members
SUB-AGENT AND MONITORING		
CMS evaluator	Predicted ÷ actual by condition group	

MY CONTRIBUTION

AI architect and builder

1	Wrote v1 by hand	DONE
Member-risk scripts for every stage: features, training, MLflow tracking and deployment.		
2	Designed the harness	DONE
The seven layers on Databricks, the four memory layers, and governance through code owners and tech-lead spec approval.		
3	Wrote the guides	DONE
The constitution, the spec templates, the global skills and the member-risk skills.		
4	Rebuilt feature creation with Spec Kit	DONE
Spec, plan, tasks and code end to end: the first stage built through the harness.		
5	Adding the agents	IN PROGRESS
The GPT code review agent and the AI Ops agent with Lakebase memory, while the other four stages move through the harness.		
6	Packaging and proving it	IN PROGRESS
The global package, the member-risk pack and a scorecard against vibe coding, ready to scale through the AIDLC champion network that has enabled 4,000+ people.		

GLOBAL

AI-DLC package

MEMBER RISK

This model only

LOCKED

Owner approval

MEMORY

Read every session

Done

In progress

What changes when feature creation runs through AI-DLC

Same brief, same data, same model, built twice. The first columns show how each approach works; the trial fills in the results.

Without AI-DLC: the risk

With AI-DLC: the control

Result pending the trial

Control still being built

THE TRIAL			
Feature creation, built twice			
Same brief, 1% sample and model			
Sealed acceptance tests			
5 planted defects			
2 change requests			
Scored by a third person			
Results fill the last column			
DIMENSION			MEASURED IN THE TRIAL
	WITHOUT AI-DLC vibe coding	WITH AI-DLC spec, skills, agents, gates	RESULTS PENDING
SECURITY PHI and access	Nothing stops PHI from reaching prompts, logs or outputs, and the agent can write anywhere the engineer can.	A hook blocks PHI in logs and outputs, a <u>guardrail agent</u> checks every PR, and agents write only to the dev catalog.	PHI exposures VIBE __ AI-DLC __ Also: blocked attempts, writes outside the dev catalog
CORRECTNESS Feature quality	No acceptance criteria up front, so leakage, duplicate claims and mixed HCC versions can pass unnoticed.	The spec sets acceptance criteria first; the golden test, leakage tests and <u>review agent</u> run before merge.	Planted defects caught VIBE __ of 5 AI-DLC __ of 5 Also: first-pass acceptance, golden-test match on 10 members
MODEL IMPACT Does it hold up	Leaky features inflate validation scores, then the model falls short on the unseen test year.	Leak-free features and a locked evaluation, so the score you see is the score that holds.	Test-year R², same model on each feature set VIBE 0. __ AI-DLC 0. __ Also: calibration by decile, gap to CMS-HCC
SPEED Time to done	A quick first draft, then rework: re-prompting, debugging and re-running by hand.	A slower start while the spec is written, then fewer rework rounds and faster changes.	Hours to done, including 2 changes VIBE __ h AI-DLC __ h Also: rework rounds, time to first passing build
COST Tokens and compute	Context is re-explained in every prompt, and failed attempts burn tokens and cluster time.	Skills load only when needed, one module per session, and fewer failed runs.	Cost per feature VIBE \$ __ AI-DLC \$ __ Also: tokens per feature, DBUs per full-data run
AUDIT Governance	Decisions live in chat history, with no named approver and no link from code back to a requirement.	The tech lead approves the spec, a code owner approves the PR, and every agent step is traced.	Actions in the audit trail VIBE __% AI-DLC __% Also: features traced to a spec, named approvals
CHANGE AND REUSE Scale	Every change starts from a fresh prompt; the blast radius is unknown and the knowledge stays with one person.	A change is a spec update that reruns from the earliest stage, and the package works for any team.	Hours per change request VIBE __ h AI-DLC __ h Also: files touched, regressions, hand-off time

A first draft is often quicker without AI-DLC. **The trial scores the whole cycle, including two change requests, because that is where rework, leakage and audit gaps cost the most.**